

Activity_Discover what is in your dataset

April 1, 2026

1 Activity: Discover what is in your dataset

1.1 Introduction

In this activity, you will discover characteristics of a dataset and use visualizations to analyze the data. This will develop and strengthen your skills in **exploratory data analysis (EDA)** and your knowledge of functions that allow you to explore and visualize data.

EDA is an essential process in a data science workflow. As a data professional, you will need to conduct this process to better understand the data at hand and determine how it can be used to solve the problem you want to address. This activity will give you an opportunity to practice that process and prepare you for EDA in future projects.

In this activity, you are a member of an analytics team that provides insights to an investing firm. To help them decide which companies to invest in next, the firm wants insights into **unicorn companies**—companies that are valued at over one billion dollars. The data you will use for this task provides information on over 1,000 unicorn companies, including their industry, country, year founded, and select investors. You will use this information to gain insights into how and when companies reach this prestigious milestone and to make recommendations for next steps to the investing firm.

1.2 Step 1: Imports

1.2.1 Import libraries and packages

First, import relevant Python libraries and modules. Use the `pandas` library and the `matplotlib.pyplot` module.

```
[2]: # Import libraries and packages
import pandas as pd
import matplotlib.pyplot as plt

### YOUR CODE HERE ###
```

1.2.2 Load the dataset into a DataFrame

The dataset provided is in the form of a csv file named `Unicorn_Companies.csv` and contains a subset of data on unicorn companies. As shown in this cell, the dataset has been automatically loaded in for you. You do not need to download the .csv file, or provide more code, in order to access the dataset and proceed with this lab. Please continue with this activity by completing the following instructions.

```
[3]: # RUN THIS CELL TO IMPORT YOUR DATA.

### YOUR CODE HERE ###
companies = pd.read_csv("Unicorn_Companies.csv")
```

Hint 1

Refer to what you learned about [loading data](#) in Python.

Hint 2

Use the function in the `pandas` library that allows you to read data from a csv file and load the data into a DataFrame.

Hint 3

Use the `read_csv()` function from the `pandas` library.

1.3 Step 2: Data exploration

1.3.1 Display the first 10 rows of the data

Next, explore the dataset and answer questions to guide your exploration and analysis of the data. To begin, display the first 10 rows of the data to get an understanding of how the dataset is structured.

```
[4]: # Display the first 10 rows of the data

### YOUR CODE HERE ###
companies.head(10)
```

```
[4]:
```

	Company	Valuation	Date Joined	Industry \
0	Bytedance	\$180B	4/7/17	Artificial intelligence
1	SpaceX	\$100B	12/1/12	Other
2	SHEIN	\$100B	7/3/18	E-commerce & direct-to-consumer
3	Stripe	\$95B	1/23/14	Fintech
4	Klarna	\$46B	12/12/11	Fintech
5	Canva	\$40B	1/8/18	Internet software & services
6	Checkout.com	\$40B	5/2/19	Fintech
7	Instacart	\$39B	12/30/14	Supply chain, logistics, & delivery
8	JUUL Labs	\$38B	12/20/17	Consumer & retail
9	Databricks	\$38B	2/5/19	Data management & analytics

	City	Country/Region	Continent	Year Founded	Funding	\
0	Beijing	China	Asia	2012	\$8B	
1	Hawthorne	United States	North America	2002	\$7B	
2	Shenzhen	China	Asia	2008	\$2B	
3	San Francisco	United States	North America	2010	\$2B	
4	Stockholm	Sweden	Europe	2005	\$4B	
5	Surry Hills	Australia	Oceania	2012	\$572M	
6	London	United Kingdom	Europe	2012	\$2B	
7	San Francisco	United States	North America	2012	\$3B	
8	San Francisco	United States	North America	2015	\$14B	
9	San Francisco	United States	North America	2013	\$3B	

Select Investors

0	Sequoia Capital China, SIG Asia Investments, S...
1	Founders Fund, Draper Fisher Jurvetson, Rothen...
2	Tiger Global Management, Sequoia Capital China...
3	Khosla Ventures, LowercaseCapital, capitalG
4	Institutional Venture Partners, Sequoia Capita...
5	Sequoia Capital China, Blackbird Ventures, Mat...
6	Tiger Global Management, Insight Partners, DST...
7	Khosla Ventures, Kleiner Perkins Caufield & By...
8	Tiger Global Management
9	Andreessen Horowitz, New Enterprise Associates...

Hint 1

Refer to [the content about exploratory data analysis in Python](#).

Hint 2

Use the function in the `pandas` library that allows you to get a specific number of rows from the top of a `DataFrame`.

Hint 3

Use the `head()` function from the `pandas` library.

Question: What do you think the “Date Joined” column represents?

I think it represents the combined date, month, and year that a given company became a “unicorn”.

Question: What do you think the “Select Investors” column represents?

I think ‘Select Investors’ represents the investors that have invested into a given company.

1.3.2 Assess the size of the dataset

Get a sense of how large the dataset is. The `size` property that `DataFrames` have can help.

```
[6]: # How large the dataset is

### YOUR CODE HERE ###
companies.size
```

```
[6]: 10740
```

Question: What do you notice about the size of the dataset?

There are 1074 rows in the dataset.

1.3.3 Determine the shape of the dataset

Identify the number of rows and columns in the dataset. The `shape` property that DataFrames have can help.

```
[7]: # Shape of the dataset

### YOUR CODE HERE ###
companies.shape
```

```
[7]: (1074, 10)
```

Question: What do you notice about the shape of the dataset?

There are 1074 rows and 10 columns in the dataset.

1.3.4 Get basic information about the dataset

To further understand what the dataset entails, get basic information about the dataset, including the data type of values in each column. There is more than one way to approach this task. In this instance, use the `info()` function from `pandas`.

```
[8]: # Get information

### YOUR CODE HERE ###
companies.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1074 entries, 0 to 1073
Data columns (total 10 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Company               1074 non-null   object
 1   Valuation             1074 non-null   object
 2   Date Joined           1074 non-null   object
 3   Industry              1074 non-null   object
 4   City                  1058 non-null   object
```

```

5   Country/Region    1074 non-null   object
6   Continent         1074 non-null   object
7   Year Founded      1074 non-null   int64
8   Funding           1074 non-null   object
9   Select Investors  1073 non-null   object
dtypes: int64(1), object(9)
memory usage: 84.0+ KB

```

Question: What do you notice about the type of data in the **Year Founded** column? Refer to the output from using `info()` above. Knowing the data type of this variable is helpful because it indicates what types of analysis can be done with that variable, how it can be aggregated with other variables, and so on.

The type of data in the ‘Year Founded’ column is 64-bit Integer.

Question: What do you notice about the type of data in the **Date Joined** column? Refer to the output from using `info()` above. Knowing the data type of this variable is helpful because it indicates what types of analysis can be done with that variable and how the variable can be transformed to suit specific tasks.

The data type of the ‘Date Joined’ column is with string or mixed-data type. It will be best to restructure this column by breaking it into date, month, and year for clearer data manipulation.

1.4 Step 3: Statistical tests

1.4.1 Find descriptive statistics

Find descriptive statistics and structure your dataset. The `describe()` function from the `pandas` library can help. This function generates statistics for the numeric columns in a dataset.

```

[9]: # Get descriptive statistics

    ### YOUR CODE HERE ###
    companies.describe()

```

```

[9]:      Year Founded
count    1074.000000
mean     2012.895717
std        5.698573
min       1919.000000
25%       2011.000000
50%       2014.000000
75%       2016.000000
max       2021.000000

```

Question: Based on the table of descriptive stats generated above, what do you notice about the minimum value in the **Year Founded** column? This is important to know because it helps you understand how early the entries in the data begin.

The minimum value in the ‘Year Founded’ column, per the table of descriptive stats generated above, shows that the earliest a company in the dataset was founded in 1919.

Question: What do you notice about the maximum value in the **Year Founded** column? This is important to know because it helps you understand the most recent year captured by the data.

The maximum value in the ‘Year Founded’ column, per the table of descriptive stats generated above, shows that the latest a company in the dataset was founded in 2021.

1.4.2 Convert the Date Joined column to datetime

Use the `to_datetime()` function from the `pandas` library to convert the **Date Joined** column to datetime. This splits each value into year, month, and date components. This is an important step in data cleaning, as it makes the data in this column easier to use in tasks you may encounter. To name a few examples, you may need to compare “date joined” between companies or determine how long it took a company to become a unicorn. Having “date joined” in datetime form would help you complete such tasks.

```
[12]: # Step 1: Use pd.to_datetime() to convert Date Joined column to datetime
      # Step 2: Update the column with the converted values

      ### YOUR CODE HERE ###
      companies['Date Joined'] = pd.to_datetime(companies['datetime'])
```

```
[13]: # Use .info() to confirm that the update actually took place

      ### YOUR CODE HERE ###
      companies.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1074 entries, 0 to 1073
Data columns (total 11 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Company               1074 non-null  object
 1   Valuation             1074 non-null  object
 2   Date Joined           1074 non-null  datetime64[ns]
 3   Industry              1074 non-null  object
 4   City                  1058 non-null  object
 5   Country/Region        1074 non-null  object
 6   Continent             1074 non-null  object
 7   Year Founded          1074 non-null  int64
 8   Funding               1074 non-null  object
 9   Select Investors      1073 non-null  object
10   datetime              1074 non-null  datetime64[ns]
dtypes: datetime64[ns](2), int64(1), object(8)
memory usage: 92.4+ KB
```

1.4.3 Create a Year Joined column

It is common to encounter situations where you will need to compare the year joined with the year founded. The `Date Joined` column does not just have year—it has the year, month, and date. Extract the year component from the `Date Joined` column and add those year components into a new column to keep track of each company's year joined.

```
[14]: # Step 1: Use .dt.year to extract year component from Date Joined column
      # Step 2: Add the result as a new column named Year Joined to the DataFrame
```

```
### YOUR CODE HERE ###
```

```
companies['Year Joined'] = companies['Date Joined'].dt.year
```

```
[41]: # Use .head() to confirm that the new column did get added
```

```
### YOUR CODE HERE ###
```

```
companies.head()
```

```
[41]:
```

	Company	Valuation	Date Joined	Industry
0	Bytedance	\$180B	2017-04-07	Artificial intelligence
1	SpaceX	\$100B	2012-12-01	Other
2	SHEIN	\$100B	2018-07-03	E-commerce & direct-to-consumer
3	Stripe	\$95B	2014-01-23	Fintech
4	Klarna	\$46B	2011-12-12	Fintech

	City	Country/Region	Continent	Year Founded	Funding
0	Beijing	China	Asia	2012	\$8B
1	Hawthorne	United States	North America	2002	\$7B
2	Shenzhen	China	Asia	2008	\$2B
3	San Francisco	United States	North America	2010	\$2B
4	Stockholm	Sweden	Europe	2005	\$4B

	Select Investors	datetime	Year Joined
0	Sequoia Capital China, SIG Asia Investments, S...	2017-04-07	2017
1	Founders Fund, Draper Fisher Jurvetson, Rothen...	2012-12-01	2012
2	Tiger Global Management, Sequoia Capital China...	2018-07-03	2018
3	Khosla Ventures, LowercaseCapital, capitalG	2014-01-23	2014
4	Institutional Venture Partners, Sequoia Capita...	2011-12-12	2011

	years_to_unicorn_status
0	5
1	10
2	10
3	4
4	6

1.5 Step 4: Results and evaluation

1.5.1 Take a sample of the data

It is not necessary to take a sample of the data in order to conduct the visualizations and EDA that follow. But you may encounter scenarios in the future where you will need to take a sample of the data due to time and resource limitations. For the purpose of developing your skills around sampling, take a sample of the data and work with that sample for the next steps of analysis you want to conduct. Use the `sample()` function for this task.

- Use `sample()` with the `n` parameter set to 50 to randomly sample 50 unicorn companies from the data. Be sure to specify the `random_state` parameter to ensure reproducibility of your work. Save the result to a variable called `companies_sampled`.

```
[57]: # Sample the data

### YOUR CODE HERE ###
companies_sample = companies.sample(n=50, random_state = 43)
companies_sample.head()
```

```
[57]:      Company Valuation Date Joined      Industry \
834   Sunbit      $1B  2021-05-20      Fintech
143   Dataiku      $5B  2019-12-04  Internet software & services
27    OpenSea     $13B  2021-07-20  E-commerce & direct-to-consumer
288   Ethos       $3B  2021-05-11      Fintech
867   Assent      $1B  2022-01-06  Supply chain, logistics, & delivery

      City Country/Region      Continent Year Founded Funding \
834   Los Angeles  United States  North America      2016   $210M
143    New York   United States  North America      2013   $647M
27    New York   United States  North America      2017   $427M
288  San Francisco  United States  North America      2016   $407M
867    Ottawa      Canada  North America      2010   $534M

      Select Investors      datetime \
834      Zeev Ventures, Group11, Chicago Ventures 2021-05-20
143      Alven Capital, FirstMark Capital, capitalG 2019-12-04
27   Andreessen Horowitz, Thirty Five Ventures, Sou... 2021-07-20
288      Sequoia Capital, Google Ventures, Accel 2021-05-11
867   Vista Equity Partners, Warburg Pincus, First A... 2022-01-06

      Year Joined  years_to_unicorn_status
834      2021      5
143      2019      6
27      2021      4
288      2021      5
867      2022     12
```


1.5.2 Visualize the time it took companies to reach unicorn status

Visualize the longest time it took companies to reach unicorn status for each industry represented in the sample. To create a bar plot to visualize this, use the `bar()` function from the `matplotlib.pyplot` module. You'll first need to prepare the data.

```
[58]: # Prepare data for plotting

#### YOUR CODE HERE ####
companies_sample["years_to_unicorn_status"] = companies_sample["Year Joined"] -
    ↪ companies_sample["Year Founded"]
```

```
[59]: # Create bar plot
# with the various industries as the categories of the bars
# and the time it took to reach unicorn status as the height of the bars

#### YOUR CODE HERE ####
plt.bar(companies_sample["Industry"],
    ↪ companies_sample["years_to_unicorn_status"])

# Set title

#### YOUR CODE HERE ####
plt.title("Time to Reach Unicorn Status by Industry (Sample Dataset)")

# Set x-axis label

#### YOUR CODE HERE ####
plt.xlabel("Industry")

# Set y-axis label

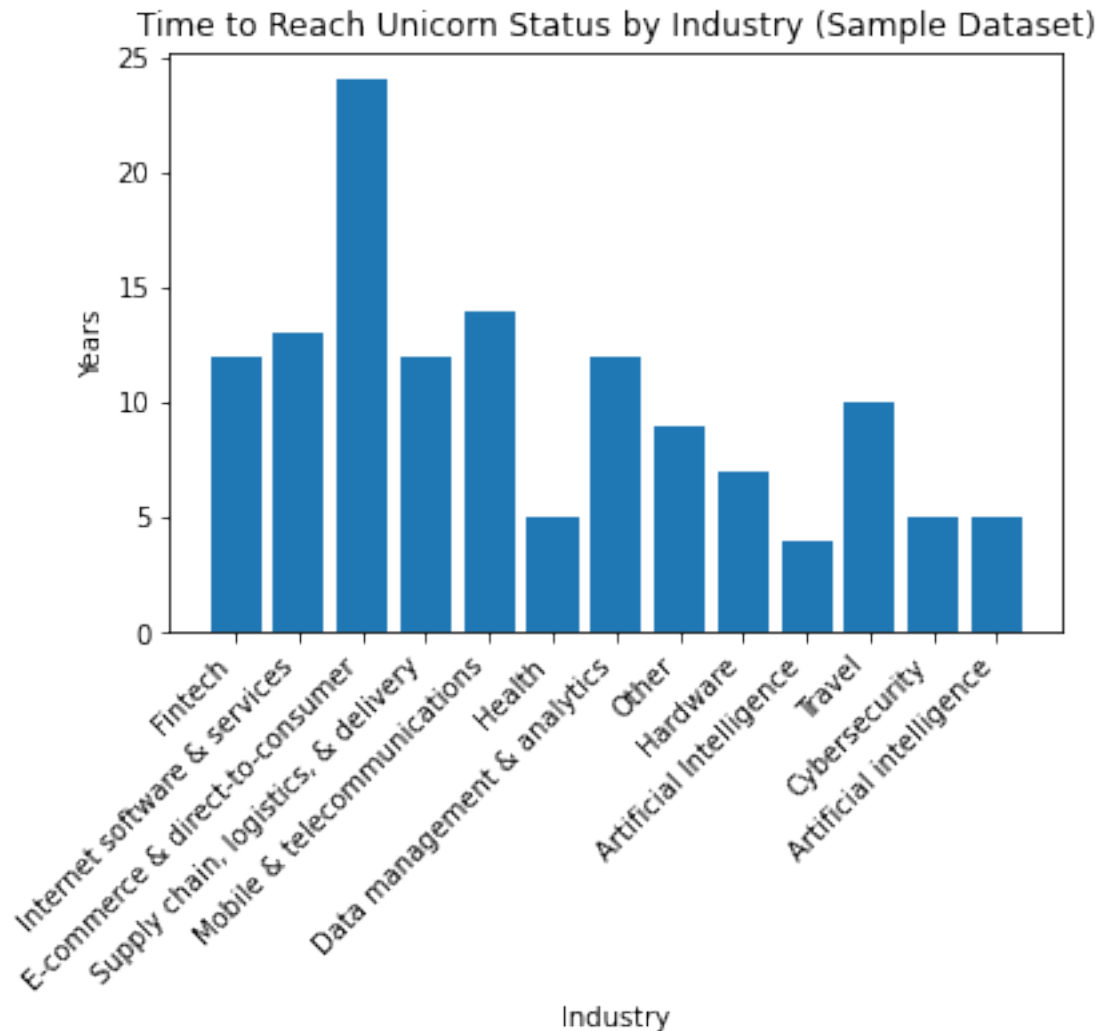
#### YOUR CODE HERE ####
plt.ylabel("Years")

# Rotate labels on the x-axis as a way to avoid overlap in the positions of the
    ↪ text

#### YOUR CODE HERE ####
plt.xticks(rotation=45, horizontalalignment = 'right')

# Display the plot

#### YOUR CODE HERE ####
plt.show()
```



Hint 1

To prepare the data for modeling, begin by creating a column that represents the number of years it took each company to reach unicorn status.

Hint 2

To prepare the data for modeling, group the dataframe by industry and get the maximum value in the newly created column for each industry.

Hint 3

You can use the `plt.bar()` to create the bar plot, passing in the categories and heights of the bars.

You can use `plt.title()`, `plt.xlabel()`, and `plt.ylabel()` to set the title, x-axis label, and y-axis label, respectively.

You can use `plt.xticks()` to rotate labels on the x-axis of a plot. The parameters `rotation=45`, `horizontalalignment='right'` can be passed in to rotate the labels by 45 degrees and align the

labels to the right.

You can use `plt.show()` to display a plot.

Question: What do you observe from this bar plot?

The bar plot shows that for this sample dataset of unicorn companies, the largest value for time taken to become a unicorn company was in the E-commerce & direct-to-consumer industry and the smallest value was in the Artificial Intelligence industry.

1.5.3 Visualize the maximum unicorn company valuation per industry

Visualize unicorn companies' maximum valuation for each industry represented in the sample. To create a bar plot to visualize this, use the `bar()` function from the `matplotlib.pyplot` module. Before plotting, create a new column that represents the companies' valuations as numbers (instead of strings, as they're currently represented). Then, use this new column to plot your data.

```
[60]: # Create a column representing company valuation as numeric data
companies_sample['valuation_in_billions'] = companies_sample['Valuation']
companies_sample['valuation_in_billions'] =_
    ↳companies_sample['valuation_in_billions'].str.replace('$', "")
companies_sample['valuation_in_billions'] =_
    ↳companies_sample['valuation_in_billions'].str.replace('B', "")
companies_sample['valuation_in_billions'] =_
    ↳companies_sample['valuation_in_billions'].astype('int')
companies_sample.head()

unicorn_industry = (companies_sample[['Industry', 'valuation_in_billions']]
                    .groupby('Industry')
                    .max()
                    .sort_values(by='valuation_in_billions'))

unicorn_industry
```

```
[60]:
```

	valuation_in_billions
Industry	
Artificial intelligence	1
Cybersecurity	1
Data management & analytics	1
Mobile & telecommunications	1
Other	1
Travel	1
Artificial Intelligence	2
Hardware	2
Health	2
Supply chain, logistics, & delivery	2
Fintech	4
Internet software & services	5

```
[61]: # Create bar plot
# with the various industries as the categories of the bars
# and the maximum valuation for each industry as the height of the bars

### YOUR CODE HERE ###
plt.bar(unicorn_industry.index,unicorn_industry['valuation_in_billions'])

# Set title

### YOUR CODE HERE ###
plt.title("Maximum Unicorn Company Valuation per Industry")

# Set x-axis label

### YOUR CODE HERE ###
plt.xlabel("Industry")

# Set y-axis label

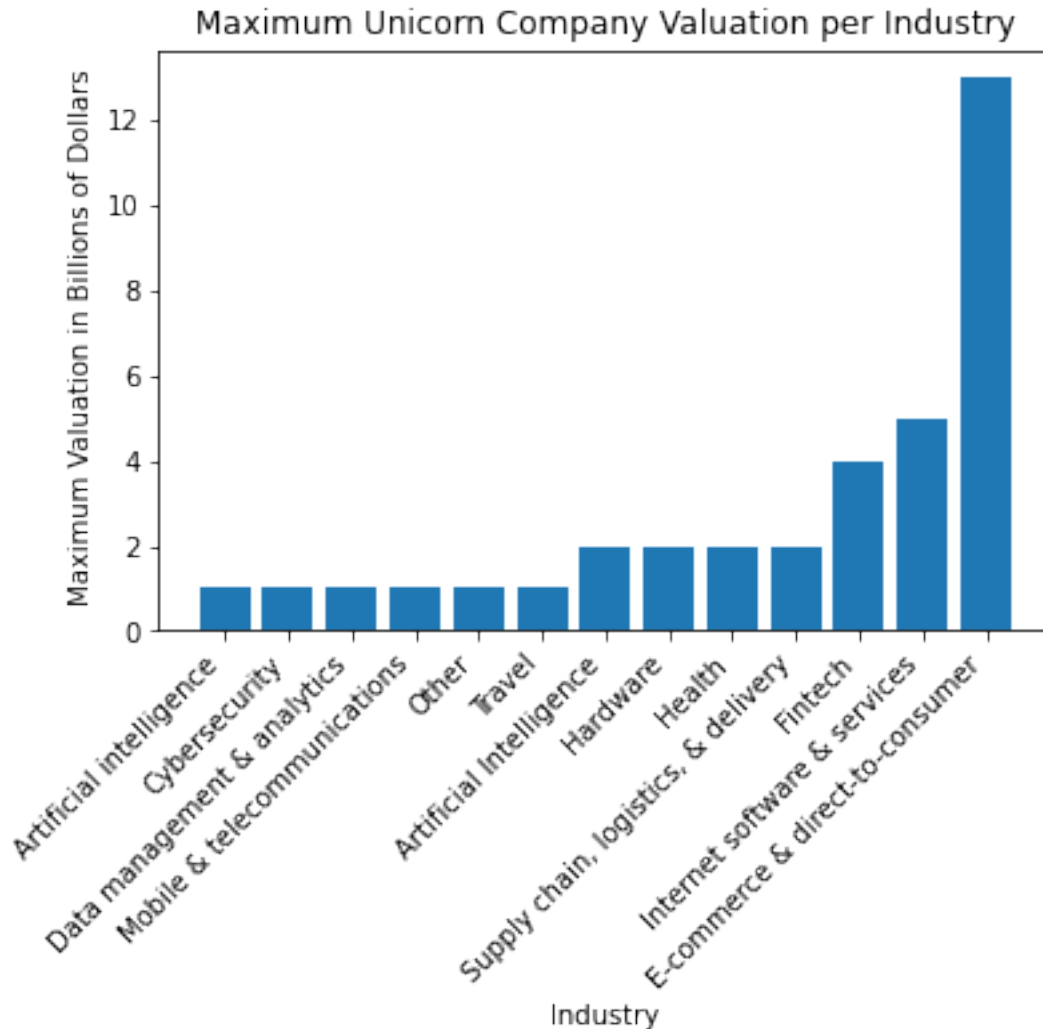
### YOUR CODE HERE ###
plt.ylabel(" Maximum Valuation in Billions of Dollars")

# Rotate labels on the x-axis as a way to avoid overlap in the positions of the
↳ text

### YOUR CODE HERE ###
plt.xticks(rotation=45, horizontalalignment='right')

# Display the plot

### YOUR CODE HERE ###
plt.show()
```



Hint 1

Repeat the process from the last task, only this time with different variables.

Hint 2

Use the function in the `matplotlib.pyplot` module that allows you to create a bar plot, specifying the category and height for each bar.

Use the functions in the `matplotlib.pyplot` module that allow you to set the title, x-axis label, and y-axis label of plots. In that module, there are also functions for rotating the labels on the x-axis and displaying the plot.

Hint 3

Use the `plt.bar()` to create the bar plot, passing in the categories and heights of the bars.

Use `plt.title()`, `plt.xlabel()`, and `plt.ylabel()` to set the title, x-axis label, and y-axis label, respectively.

Use `plt.xticks()` to rotate labels on the x-axis of a plot. The parameters `rotation=45`, `horizontalalignment='right'` can be passed in to rotate the labels by 45 degrees and align the labels to the right.

Use `plt.show()` to display a plot.

Question: What do you observe from this bar plot?

This plot shows that for this sample of the dataset, the highest maximum valuation occurred in the E-commerce & direct-to-consumer industry, and the lowest maximum valuation occurred in the Artificial Intelligence, Cybersecurity, Data management & analytics, Mobile & telecommunication, Other, and Travel industries.

1.6 Considerations

What are some key takeaways that you learned from this lab?

I learned how helpful it is to structure the data as needed to answer the questions asked. I also observed that the charts for some samples look very different. So, it is important to provide 'random_state' to the sample for the reproducibility of my work.

What findings would you share with others?

I would further explore some more samples and share the industries that have faster turnaround for the companies to reach unicorn state. This would be valuable information for the stakeholders in this project as it would help them determine their future investments.

What recommendations would you share with stakeholders based on these findings?

I would share my findings on the industries that have companies that take the shortest amount of time to reach unicorn state to help the stakeholders make decisions for their future investments.

References

Bhat, M.A. (2022, March). *Unicorn Companies*.

Congratulations! You've completed this lab. However, you may not notice a green check mark next to this item on Coursera's platform. Please continue your progress regardless of the check mark. Just click on the "save" icon at the top of this notebook to ensure your work has been logged.